Pip: A Lightweight AI Assistant Window with Safe Local Tool Access

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Abstract

A student’s AI chat can answer questions about the world but cannot open a single file on their own machine. Everything must be pasted in, and the useful automation — an assistant that reads your notes, finds your document, saves what it worked out — is reserved for users with a paid API key or a capable local machine. The capability gap is a pricing gap, and it lands on exactly the users who would benefit most.

Pip is a small always-on-top desktop window, built with Tauri and TypeScript, that sits beside the AI chat service a user already has and gives it hands. A local Model Context Protocol server exposes seven tools — list a folder, read a document, search notes, write a note, render a digest or a quiz — guarded by a folder allowlist with path canonicalisation, native confirmation before any write, authenticated connections, output size limits and an append-only audit log, while the chat webview is deliberately granted no application permissions. One of those seven tools converts PDF, DOCX, PPTX and XLSX using Microsoft’s MarkItDown; the window also carries a Pomodoro timer and quick notes, because a tool that is not pleasant to keep open does not get used.

Pip needs no API key and no local model. A free Claude account reached the local tool server through a public tunnel and executed calls end to end, and adversarial testing closed path traversal and unauthenticated access while exposing an approval bypass that a green test suite had missed. The document tool was measured and turned out not to save tokens at all — conversion makes a prompt 12% larger — which is reported as a negative result rather than presented as a feature.

Keywords: AI assistant, Model Context Protocol, local tool access, permission boundary, Tauri